

When Are Two Things Equal?

Math on Tap at Mother Road Brewery

Dana C. Ernst

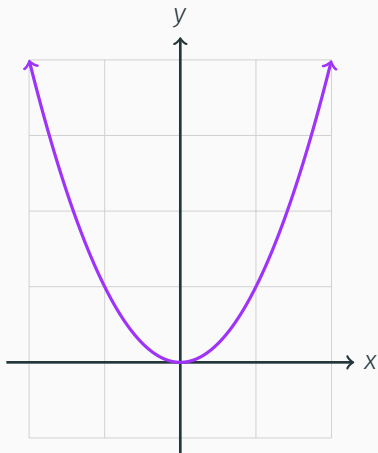
Department of Mathematics and Statistics

Northern Arizona University

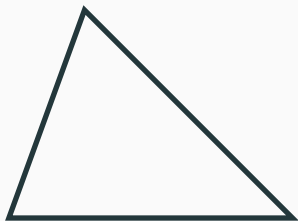
September 2, 2026

When Are Two Things Equal?

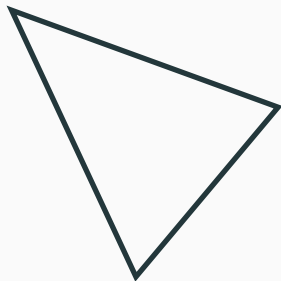
$$\boxed{f(x) = x^2} \stackrel{?}{=}$$



When Are Two Things Equal?



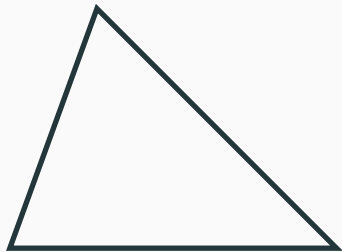
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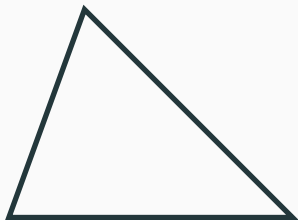
When Are Two Things Equal?



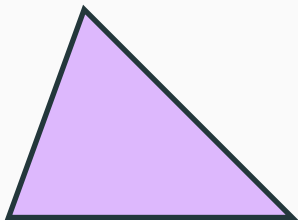
$\stackrel{?}{=}$



When Are Two Things Equal?



$\overset{?}{=}$



When Are Two Things Equal?

$$\frac{1}{3} \stackrel{?}{=} \frac{4}{12}$$

$$\frac{1}{3} \stackrel{?}{=} 0.333 \dots$$

$$1 \stackrel{?}{=} 0.999 \dots$$

When Are Two Things Equal?

6 $\stackrel{?}{=}$ 6

6 $\stackrel{?}{=}$ six

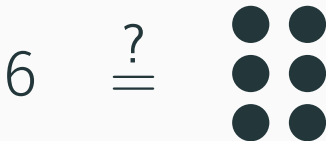
6 $\stackrel{?}{=}$ 6

six $\stackrel{?}{=}$ seis

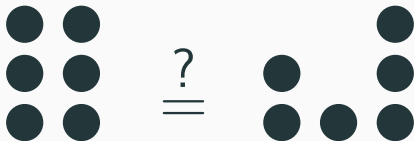
6 $\stackrel{?}{=}$ 6

6 $\stackrel{?}{=}$ x

When Are Two Things Equal?



When Are Two Things Equal?

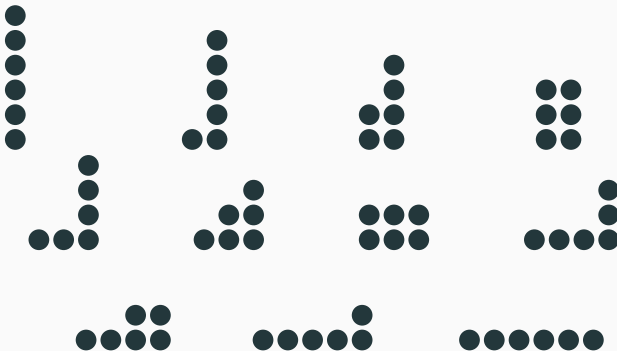


When Are Two Things Equal?



When Are Two Things Equal?

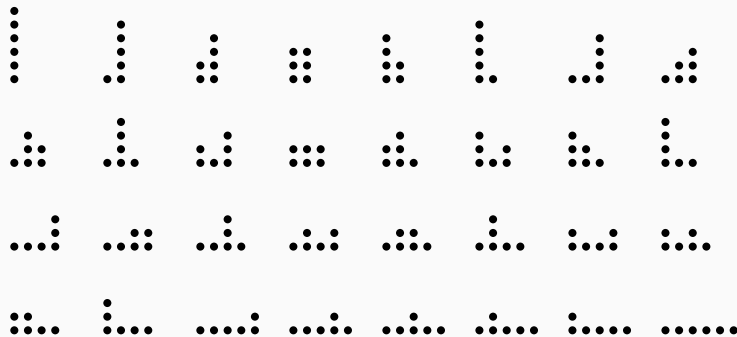
11 integer partitions of 6



No known “nice” formula for number of
integer partitions of n

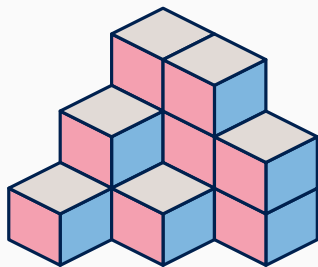
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32 ordered integer partitions of 6



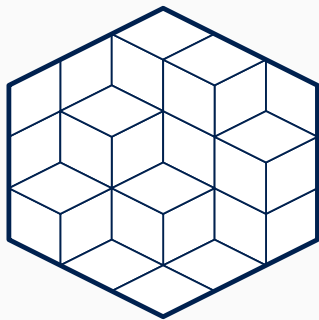
There are 2^{n-1} ordered integer partitions of n

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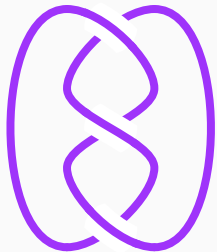


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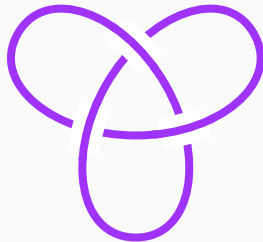
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When Are Two Things Equal?



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When Are Two Things Equal?

$$\{C, A, T\} \stackrel{?}{=} \{T, C, A\}$$

When Are Two Things Equal?

$$\{2, 3, 5, 8\} \stackrel{?}{=} 01101001$$

When Are Two Things Equal?

16 subsets / binary strings for $n = 4$

$\emptyset$: 0000 $\{4\}$: 0001 $\{3\}$: 0010 $\{3, 4\}$: 0011

$\{2\}$: 0100 $\{2, 4\}$: 0101 $\{2, 3\}$: 0110 $\{2, 3, 4\}$: 0111

$\{1\}$: 1000 $\{1, 4\}$: 1001 $\{1, 3\}$: 1010 $\{1, 3, 4\}$: 1011

$\{1, 2\}$: 1100 $\{1, 2, 4\}$: 1101 $\{1, 2, 3\}$: 1110 $\{1, 2, 3, 4\}$: 1111

There are 2^n subsets of a set with n elements

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$$CAT \stackrel{?}{=} TCA$$

$$123 \stackrel{?}{=} ABC$$

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6 permutations of 3 objects

123



ABC

132



ACB

213



BAC

231



BCA

312



CAB

321



CBA

There are $n!$ permutations of n objects

When Are Two Things Equal?

$$\{1, 4, 5\} \mid \{2\} \mid \{3, 6\} \stackrel{?}{=} \{2\} \mid \{3, 6\} \mid \{1, 4, 5\}$$

When Are Two Things Equal?

15 set partitions for $n = 4$

$\{1, 2, 3, 4\}$	$\{1\} \mid \{2, 3, 4\}$	$\{1, 2\} \mid \{3, 4\}$
$\{1, 2, 3\} \mid \{4\}$	$\{1, 2, 4\} \mid \{3\}$	$\{1, 3\} \mid \{2, 4\}$
$\{1, 3, 4\} \mid \{2\}$	$\{1, 4\} \mid \{2, 3\}$	$\{1\} \mid \{2\} \mid \{3, 4\}$
$\{1\} \mid \{2, 3\} \mid \{4\}$	$\{1\} \mid \{2, 4\} \mid \{3\}$	$\{1, 2\} \mid \{3\} \mid \{4\}$
$\{1, 3\} \mid \{2\} \mid \{4\}$	$\{1, 4\} \mid \{2\} \mid \{3\}$	$\{1\} \mid \{2\} \mid \{3\} \mid \{4\}$

Number of set partitions for n is given
by n th Bell number

When Are Two Things Equal?

75 ordered set partitions for $n = 4$

$\{1, 2, 3, 4\}$	$\{1\} \mid \{2, 3, 4\}$	$\{2, 3, 4\} \mid \{1\}$	$\{1, 2\} \mid \{3, 4\}$	$\{3, 4\} \mid \{1, 2\}$
$\{1, 2, 3\} \mid \{4\}$	$\{4\} \mid \{1, 2, 3\}$	$\{1, 2, 4\} \mid \{3\}$	$\{3\} \mid \{1, 2, 4\}$	$\{1, 3\} \mid \{2, 4\}$
$\{2, 4\} \mid \{1, 3\}$	$\{1, 3, 4\} \mid \{2\}$	$\{2\} \mid \{1, 3, 4\}$	$\{1, 4\} \mid \{2, 3\}$	$\{2, 3\} \mid \{1, 4\}$
$\{1\} \mid \{2\} \mid \{3, 4\}$	$\{1\} \mid \{3, 4\} \mid \{2\}$	$\{2\} \mid \{1\} \mid \{3, 4\}$	$\{2\} \mid \{3, 4\} \mid \{1\}$	$\{3, 4\} \mid \{1\} \mid \{2\}$
$\{3, 4\} \mid \{2\} \mid \{1\}$	$\{1\} \mid \{2, 3\} \mid \{4\}$	$\{1\} \mid \{4\} \mid \{2, 3\}$	$\{2, 3\} \mid \{1\} \mid \{4\}$	$\{2, 3\} \mid \{4\} \mid \{1\}$
$\{4\} \mid \{1\} \mid \{2, 3\}$	$\{4\} \mid \{2, 3\} \mid \{1\}$	$\{1\} \mid \{2, 4\} \mid \{3\}$	$\{1\} \mid \{3\} \mid \{2, 4\}$	$\{2, 4\} \mid \{1\} \mid \{3\}$
$\{2, 4\} \mid \{3\} \mid \{1\}$	$\{3\} \mid \{1\} \mid \{2, 4\}$	$\{3\} \mid \{2, 4\} \mid \{1\}$	$\{1, 2\} \mid \{3\} \mid \{4\}$	$\{1, 2\} \mid \{4\} \mid \{3\}$
$\{3\} \mid \{1, 2\} \mid \{4\}$	$\{3\} \mid \{4\} \mid \{1, 2\}$	$\{4\} \mid \{1, 2\} \mid \{3\}$	$\{4\} \mid \{3\} \mid \{1, 2\}$	$\{1, 3\} \mid \{2\} \mid \{4\}$
$\{1, 3\} \mid \{4\} \mid \{2\}$	$\{2\} \mid \{1, 3\} \mid \{4\}$	$\{2\} \mid \{4\} \mid \{1, 3\}$	$\{4\} \mid \{1, 3\} \mid \{2\}$	$\{4\} \mid \{2\} \mid \{1, 3\}$
$\{1, 4\} \mid \{2\} \mid \{3\}$	$\{1, 4\} \mid \{3\} \mid \{2\}$	$\{2\} \mid \{1, 4\} \mid \{3\}$	$\{2\} \mid \{3\} \mid \{1, 4\}$	$\{3\} \mid \{1, 4\} \mid \{2\}$
$\{3\} \mid \{2\} \mid \{1, 4\}$	$\{1\} \mid \{2\} \mid \{3\} \mid \{4\}$	$\{1\} \mid \{2\} \mid \{4\} \mid \{3\}$	$\{1\} \mid \{3\} \mid \{2\} \mid \{4\}$	$\{1\} \mid \{3\} \mid \{4\} \mid \{2\}$
$\{1\} \mid \{4\} \mid \{2\} \mid \{3\}$	$\{1\} \mid \{4\} \mid \{3\} \mid \{2\}$	$\{2\} \mid \{1\} \mid \{3\} \mid \{4\}$	$\{2\} \mid \{1\} \mid \{4\} \mid \{3\}$	$\{2\} \mid \{3\} \mid \{1\} \mid \{4\}$
$\{2\} \mid \{3\} \mid \{4\} \mid \{1\}$	$\{2\} \mid \{4\} \mid \{1\} \mid \{3\}$	$\{2\} \mid \{4\} \mid \{3\} \mid \{1\}$	$\{3\} \mid \{1\} \mid \{2\} \mid \{4\}$	$\{3\} \mid \{1\} \mid \{4\} \mid \{2\}$
$\{3\} \mid \{2\} \mid \{1\} \mid \{4\}$	$\{3\} \mid \{2\} \mid \{4\} \mid \{1\}$	$\{3\} \mid \{4\} \mid \{1\} \mid \{2\}$	$\{3\} \mid \{4\} \mid \{2\} \mid \{1\}$	$\{4\} \mid \{1\} \mid \{2\} \mid \{3\}$
$\{4\} \mid \{1\} \mid \{3\} \mid \{2\}$	$\{4\} \mid \{2\} \mid \{1\} \mid \{3\}$	$\{4\} \mid \{2\} \mid \{3\} \mid \{1\}$	$\{4\} \mid \{3\} \mid \{1\} \mid \{2\}$	$\{4\} \mid \{3\} \mid \{2\} \mid \{1\}$

Number of ordered set partitions for n is
given by n th Fubini number

When Are Two Things Equal?

$$\{2\} \mid \{3, 6\} \mid \{1, 4, 5\} \stackrel{?}{=} 312332$$

When Are Two Things Equal?

13 ordered set partitions / Cayley permutations for $n = 3$

$\{1, 2, 3\}: 111$

$\{1\} \mid \{2\} \mid \{3\}: 123$

$\{1, 2\} \mid \{3\}: 112$

$\{1\} \mid \{3\} \mid \{2\}: 132$

$\{1, 3\} \mid \{2\}: 121$

$\{2\} \mid \{1\} \mid \{3\}: 213$

$\{1\} \mid \{2, 3\}: 122$

$\{3\} \mid \{1\} \mid \{2\}: 231$

$\{2, 3\} \mid \{1\}: 211$

$\{2\} \mid \{3\} \mid \{1\}: 312$

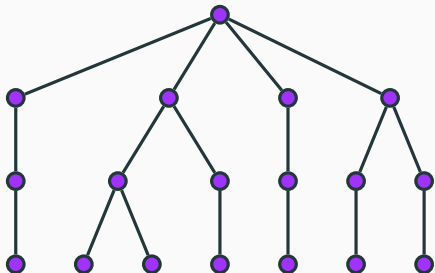
$\{2\} \mid \{1, 3\}: 212$

$\{3\} \mid \{2\} \mid \{1\}: 321$

$\{3\} \mid \{1, 2\}: 221$

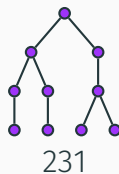
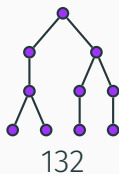
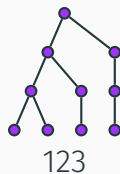
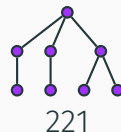
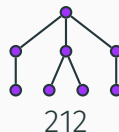
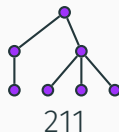
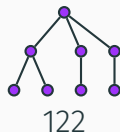
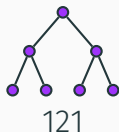
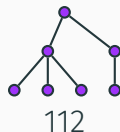
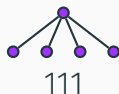
When Are Two Things Equal?

312332 $\stackrel{?}{=}$



When Are Two Things Equal?

13 Fraenkel trees for $n = 3$



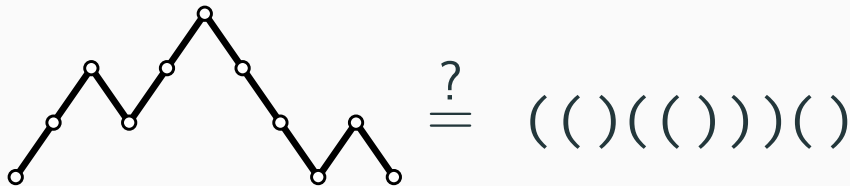
When Are Two Things Equal?

$$\frac{1}{2 - e^x} \stackrel{?}{=} \frac{1}{0!}x^0 + \frac{1}{1!}x + \frac{3}{2!}x^2 + \frac{13}{3!}x^3 + \frac{75}{4!}x^4 + \dots$$

When Are Two Things Equal?

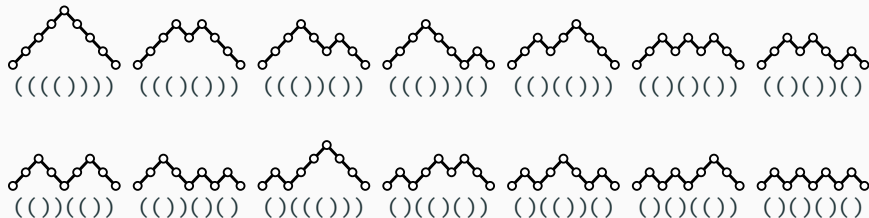
$$\frac{1}{2 - e^x} \stackrel{?}{=} (1, 1, 3, 13, 75, \dots)$$

When Are Two Things Equal?



When Are Two Things Equal?

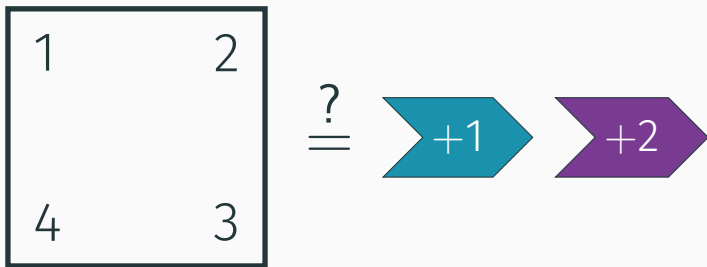
14 Dyck paths / balanced parentheses for $n = 4$



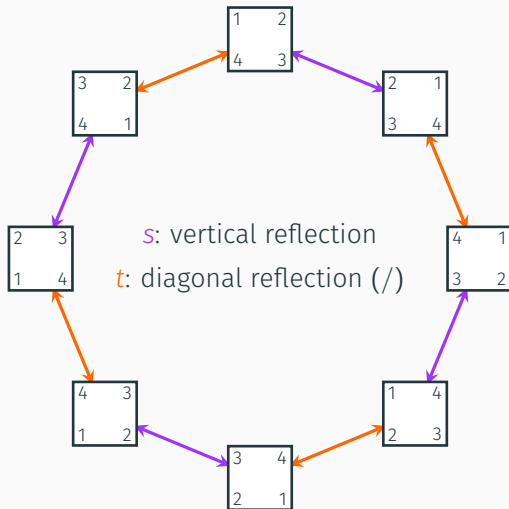
Number of Dyck paths for n is the n th Catalan number:

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

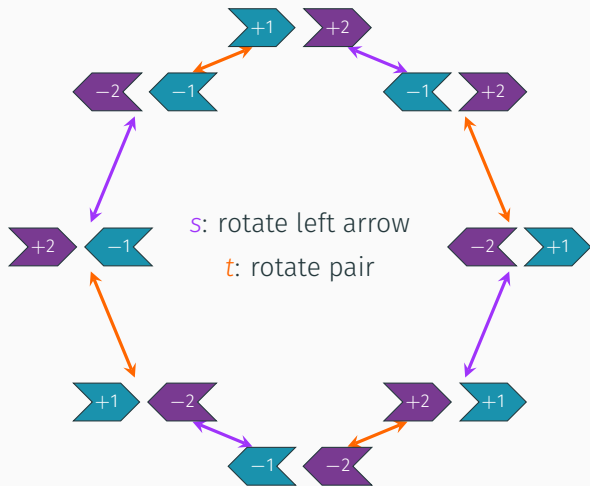
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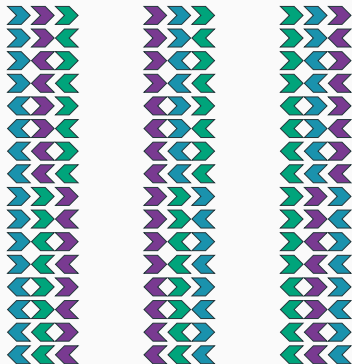
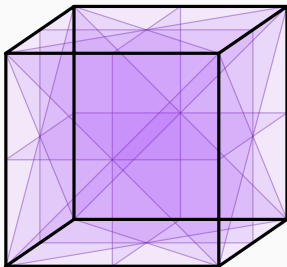


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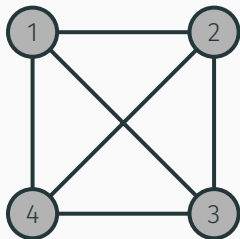
When Are Two Things Equal?

48 symmetries of cube correspond to reversals
acting on signed permutations of length 3

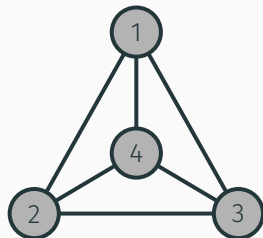


In general, the $2^n \cdot n!$ symmetries of an n -dimensional hypercube correspond to the rearrangements of a chromosome with n blocks.

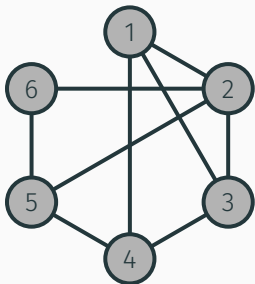
When Are Two Things Equal?



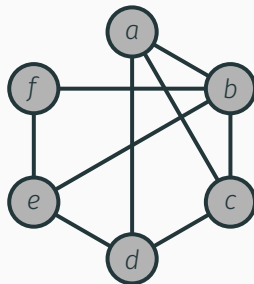
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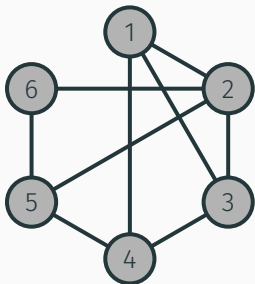
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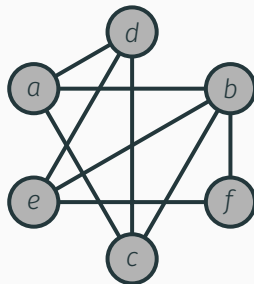
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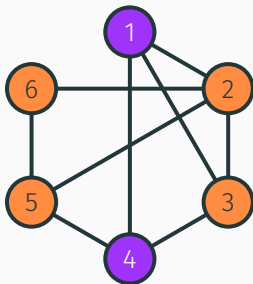
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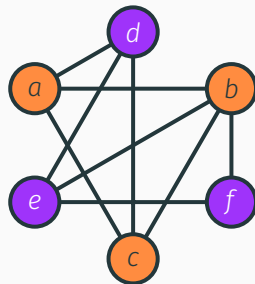
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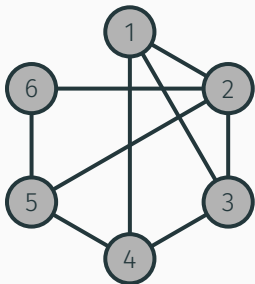
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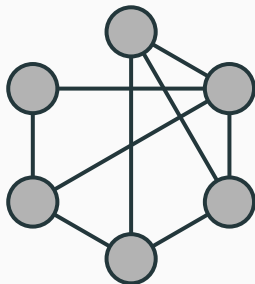
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When Are Two Things Equal?



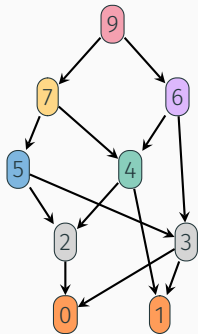
$\stackrel{?}{=}$



When Are Two Things Equal?

Subtract 2 or 3

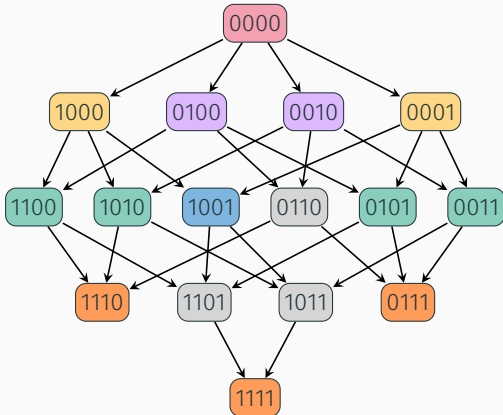
Start with 9 tokens.



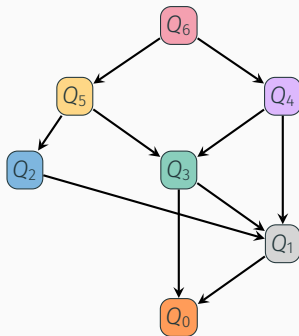
$\stackrel{?}{=}$

Treblecross on 4 Cells

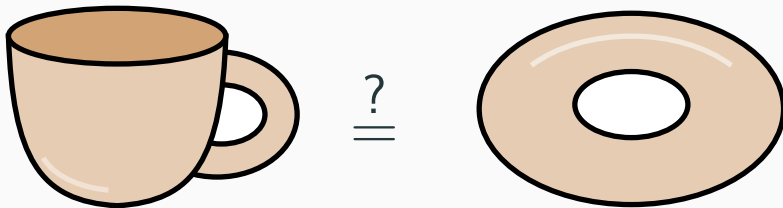
Place a 1; a run of three ends play.



When Are Two Things Equal?



When Are Two Things Equal?



[Watch: Coffee mug to doughnut](#)

Thank You!

Coming in October (possibly November):

The Congressional Apportionment Problem

Dr. Shannon Guerrero



Department of Mathematics
and Statistics



College of the Environment,
Forestry, and Natural Sciences

